SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I Shaik Ashraf/192324088 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. **Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.**

   Sol:

   <u>Type-1</u> hypervisors run directly on the physical hardware, so they provide better performance, stronger security, and easier management for large cloud environments. They have lower overhead and are more stable because there is no host operating system in between.

   <u>Type-2</u> hypervisors run on top of a host operating system. They are simple to install and useful for testing or personal use, but they consume more resources and have a higher security risk because the host operating system can also be attacked.

   For a cloud datacenter hosting mission-critical applications, Type-1 hypervisors are the better choice because they offer high performance, better isolation, improved reliability, and centralized management.

2. **A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.**

   Sol:

   Virtualization creates separate virtual machines for different workloads such as finance, healthcare, and student portals. Each VM has its own operating system, memory, storage, and applications, so one workload does not directly affect another. This improves security, privacy, and resource control.

   Two major attack surfaces are:

   • Hypervisor attacks – If the hypervisor is compromised, multiple virtual machines may be affected.

   • Virtual network attacks – Weak virtual network configuration can allow unauthorized access or data interception between virtual machines.

3. **Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.**

   Sol:

   The host shows CPU usage of 92%, memory usage of 88%, storage I/O latency of 35 ms, and VM boot time of 170 seconds. These values indicate that the server is overloaded.

   Likely bottlenecks:
   - Very high CPU utilization.
   - High memory consumption.
   - Slow storage performance causing long VM startup times.

   Corrective actions:
   - Move some virtual machines to another host using live migration.
   - Increase CPU and RAM resources if possible.
   - Upgrade storage to faster SSD or NVMe drives.
   - Remove unused virtual machines and optimize resource allocation.

4. **Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.**

   Sol:

   A Software Defined Datacenter manages compute, storage, and networking through software instead of manual hardware configuration.

   Compute virtualization allows virtual machines to be created quickly.
   Storage virtualization combines different storage devices into one flexible storage pool.
   Network virtualization creates virtual networks that can be configured without changing physical devices.

   Compared with a traditional datacenter, an SDDC provides faster deployment, easier management, better scalability, and reduced operational effort.

5. **A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.**

   Sol:

   Identity mismatches occur when different cloud providers use different authentication methods, user formats, or access policies. As a result, users may fail to log in or receive incorrect permissions.

   A secure solution is to use identity federation with standards such as SAML or OAuth/OpenID Connect. A central Identity Provider (IdP) can authenticate users once and allow secure access to multiple cloud providers. Multi-factor authentication, role-based access control, encryption, and regular auditing should also be used to improve privacy and security.